



Car Rental Backend (Flask)

This is the backend service for the **Car Rental Platform**, built with **Flask**, SQLAlchemy, and JWT authentication.

It provides APIs for user authentication, car management, bookings, and payments.



Setup Instructions

1 Clone the repository

Bash

```
git clone https://github.com/your-username/car_rental_backend.git
cd car_rental_backend
```

2 Create and activate a virtual environment

Bash

```
python -m venv venv

# On Windows (PowerShell):
.\venv\Scripts\Activate.ps1

# On Linux/Mac:
source venv/bin/activate
```

3 Install dependencies

Bash

```
pip install -r requirements.txt
```

4 Initialize the database

Bash

```
python create_db.py
```

5 Run the server

Bash

```
python manage.py
```

The server will start on:

👉 <http://127.0.0.1:5000>

🔑 Environment Variables

Copy `.env.template` to `.env` and update values:

Bash

```
cp .env.template .env
```

Example `.env` :

Plain Text

```
FLASK_ENV=development
SECRET_KEY=change_this_secret
DATABASE_URL=sqlite:///data.db
JWT_SECRET_KEY=change_this_jwt_secret
```

🐳 Run with Docker

1 Build the Docker image

Bash

```
docker build -t car_rental_backend .
```

2 Run the container

Bash

```
docker run -p 5000:5000 --env-file .env car_rental_backend
```

3 Or use Docker Compose

Bash

```
docker-compose up --build
```

Push to Docker Hub

1 Login to Docker Hub

Bash

```
docker login
```

2 Tag the image

Bash

```
docker tag car_rental_backend your-dockerhub-username/car_rental_backend:latest
```

3 Push the image

Bash

```
docker push your-dockerhub-username/car_rental_backend:latest
```

4 Run from Docker Hub (any machine)

Bash

```
docker pull your-dockerhub-username/car_rental_backend:latest
docker run -p 5000:5000 your-dockerhub-username/car_rental_backend:latest
```

Project Structure

Plain Text

```
car_rental_backend/
|— app/
|   |— __init__.py
```

```
|
|   ├── models.py
|   ├── routes/
|   |   ├── auth.py
|   |   ├── cars.py
|   |   ├── bookings.py
|   |   └── payments.py
|   └── schemas.py
|── requirements.txt
|── manage.py
|── create_db.py
|── Dockerfile
|── docker-compose.yml
|── .dockerignore
|── .env.template
|── README.md
```

GitHub Workflow

1 Initialize Git (first time only)

2 Add remote repository

```
Bash

git remote add origin https://github.com/your-username/car_rental_backend.git
```

```
git remote add origin https://github.com/your-username/car_rental_backend.git
```

3 Stage and commit changes

```
Bash

git add .
git commit -m "Initial commit"
```

```
git add .
git commit -m "Initial commit"
```

4 Push to GitHub

```
git push -u origin main
```

API Documentation (Postman)

Auth Routes

Register

POST /auth/register

Body (JSON):

JSON

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "123456",
  "role": "renter"
}
```

Login

POST /auth/login

Body (JSON):

JSON

```
{
  "email": "john@example.com",
  "password": "123456"
}
```

Response:

JSON

```
{
  "message": "Login successful",
  "access_token": "your_jwt_token",
  "user": {
    "id": 1,
    "name": "John Doe",
  }
}
```

```
    "email": "john@example.com",  
    "role": "renter"  
  }  
}
```

Car Routes

Add Car (Owner only)

POST /cars

Headers:

Plain Text

Authorization: Bearer <your_token>

Body (JSON):

JSON

```
{  
  "brand": "Toyota",  
  "model": "Corolla",  
  "year": 2022,  
  "price_per_day": 500  
}
```

List Cars

GET /cars

Booking Routes

Create Booking (Renter only)

POST /bookings

Headers:

Plain Text

Authorization: Bearer <your_token>

Body (JSON):

JSON

```
{
  "car_id": 1,
  "start_date": "2025-09-20",
  "end_date": "2025-09-25"
}
```

Payment Routes

Make Payment

POST /payments

Headers:

Plain Text

Authorization: Bearer <your_token>

Body (JSON):

JSON

```
{
  "booking_id": 1,
  "amount": 2500,
  "method": "credit_card"
}
```

Notes

- Do NOT commit `.env` or `data.db` to GitHub (already in `.gitignore`).
- Use `.env.template` as a guide for environment variables.
- Make sure Docker Desktop is running before using Docker commands.
- Always include the `Authorization` header for protected routes.